

workspaces, and hence developers do not have to write different build scripts for Windows and Linux machines. The WSL 2 backend greatly reduces the resource requirements of running a container. This is because WSL 2 makes sure that Docker Desktop uses up memory and CPU only in the required amounts. The direct advantage of this fact is that the current version of Docker Desktop brings down the Docker daemon startup time to less than 10 seconds. Adding to this, when WSL 2 is running in a Windows machine, Docker commands can even be executed directly from Windows PowerShell.

It is very easy to install and run Docker Desktop with the WSL 2 backend. First, enable WSL 2 in your machine if it is not already enabled. Then, follow the given steps.

1. Download the latest stable release of Docker Desktop for Windows from <https://hub.docker.com/editions/community/docker-ce-desktop-windows>.
2. Run the downloaded installer by double-clicking it and following the instructions. Turn on the WSL 2 feature if prompted during installation.
3. A *Start menu* entry of Docker Desktop gets created. Launch Docker Desktop from there.

4. Go to the *General* tab on the *Settings* page.
5. Select the 'Use WSL 2 based engine' checkbox if it is not selected by default, and then click on the *Apply & Restart* button.

Now, let us learn how to perform some basic operations using Docker Desktop GUI.

In the bottom-left corner, we can see the status of the Docker service. The left panel lists two tabs – *Images* and *Containers/Apps*. The *Images* tab shows various images created locally as well as in the private repository on Docker Hub. We can pull images from the Docker Hub repository by clicking on the *Pull* button that appears when we hover the mouse pointer over an image. The pulled image will then get listed in the local images tab. To run an image in a container, click on the *Run* button beside each local image. The new container window will open. Enter details like container name, port mappings, and volume mappings, and then click the *Run* button. This creates a container shown in the *Containers/Apps* tab, along with other containers created by us. Clicking on any container name shows the *Details* page of the container. This page has three options:

- **Logs:** This shows the container output and logs generated during

container execution, as shown in Figure 9.

- **Inspect:** This shows various properties of the environment setup in the container. This includes properties like environment variables, ports, and volumes bound to the container, as shown in Figure 10.
- **Stats:** This shows various container statistics like memory usage, CPU usage, data read from and written to the disk, and data sent and received over the network, as shown in Figure 11.

One of the main advantages of using Docker Desktop on top of WSL 2 is that it allows us to execute Docker commands directly from the Windows PowerShell. The following command pulls the Jenkins image from Docker Hub (if not available locally) and runs it in a container:

```
$docker run -p 8080:8080 -p 50000:50000
jenkins/jenkins:lts
```

You can save the container as an image by running the following command:

```
$docker commit <container_name>
```

```
<docker_hub_repository>/<new_image_name>:<image_tag>
```

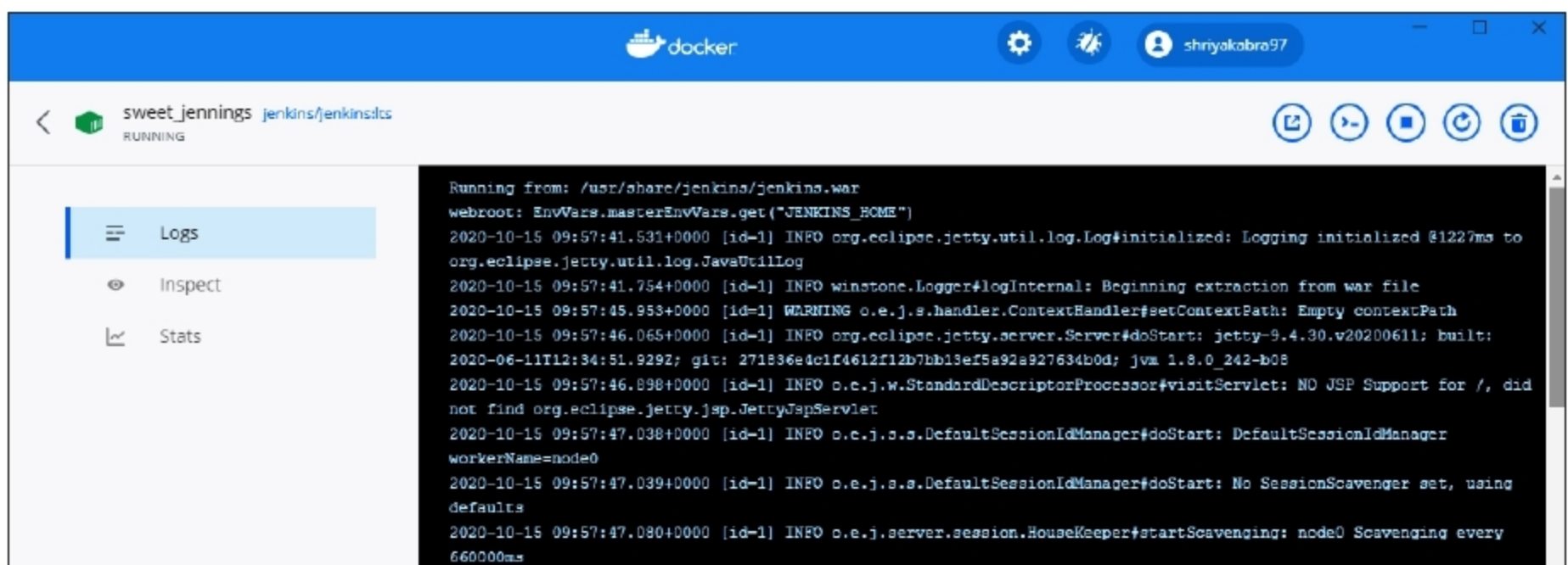


Figure 9: Viewing container logs using Docker Desktop